

8/1/24

Biology

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Page No.: 1						YOUVA
Date:						

1. P
2. Pans
3. Both 1 and 2
4. Water
5. photo phosphorylation
6. Decreased level of luteinizing hormone in the blood.
7. Woman X is conceived
8. E
9. it does not have glucose
10. Lymph
11. Centromere
12. A, B and D are correct
13. Meiosis and fertilization
14. Absent of cell sap
15. Both 1 and 2 are correct but statement 2 is not correct explanation of 1.

16.

- i) a) Monohybrid cross
- b) Amniotic fluid
- c) Paternal twins
- d) Auditory nerve
- e) Concave lens

- ii) a) True
- b) True
- c) True
- d) True
- e) False

- iii) a) blood sugar
- b) Testosterone
- c) Larynx
- d) Cranium
- e) AIDS

- iv) a) Alcohol consumption affects the cerebellum of the person, which is responsible over the controlling of skeletal muscles. Cerebellum is not able to control them and thus an alcoholic person walks clumsily.
- b) In some xerophytes leaves are reduced to spine in order to reduce the loss of water in the form of water vapours by transpiration.

T F S S
YOUVA

M T W T F S S
Page No: 3
Date: YOUVA

c) gametes must be produced by meiosis, such that each one get 23 chromosomes only, because after fertilization 46 chromosomes will be there and this number will remain constant.

d)

e) on entering the poorly lit room, a person gets feels blinded for sometime because light has divided the rodophin which requires some time for combination.

v) a) Ovulation, Fertilization, Implantation, Gestation and childbirth.

b) testes, sperm, sperm duct, semen, penis

c) ii)

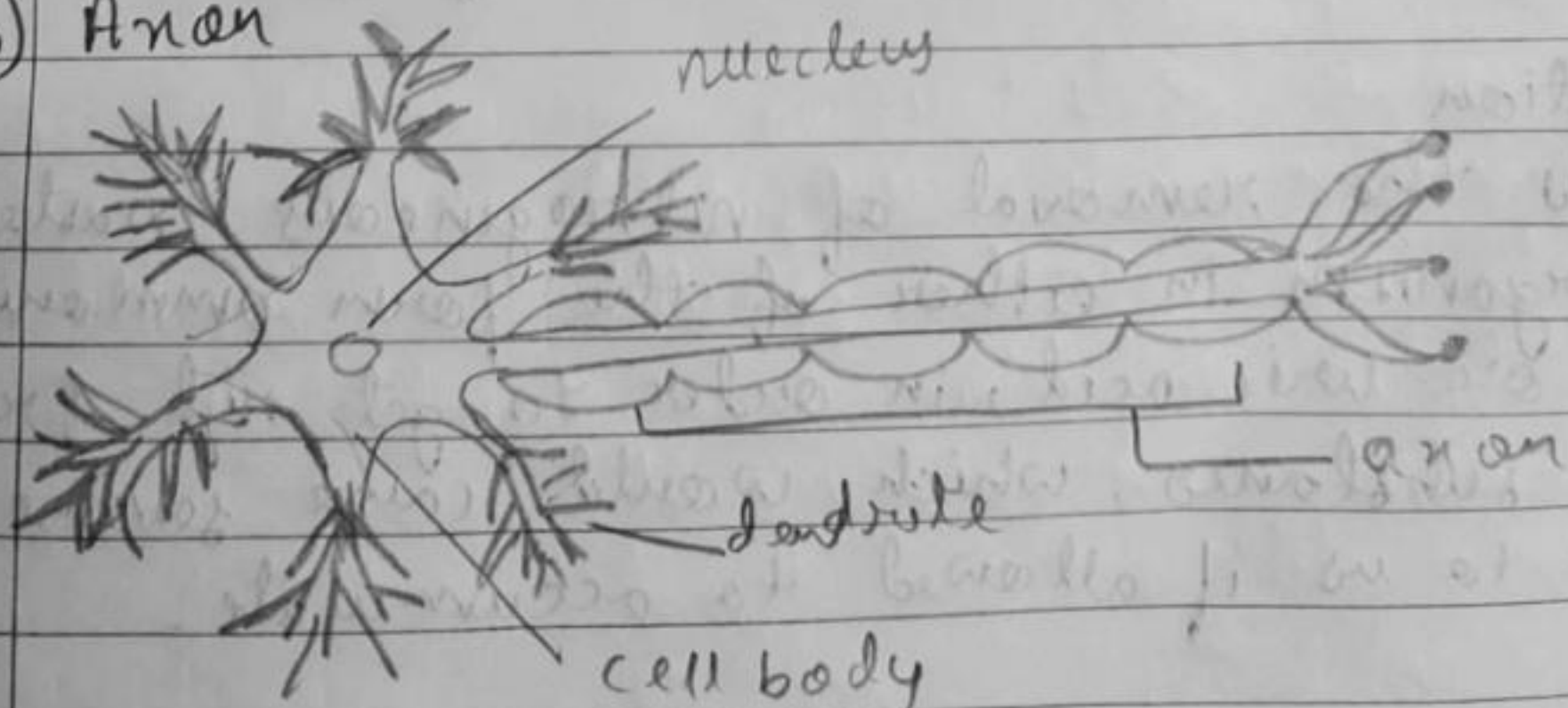
d) Homo erectus, Neanderthal man, Cro-magnon, Homo-sapien

e) Platelets, Thrombin, Fibrinogen, Fibrin

17. i) a) Dendrites

b) Axon

ii)



structure of neuron

- iii) a) Phototropism
- b) Thigmotropism

iv) a) Population
 It refers to the count of people living in a particular area and in a particular period of time. This is termed as population.

b) Tubectomy
 It is one of the mechanical methods for reducing population. It is the surgical removal of a particular part of oviduct or fallopian tubes of female.

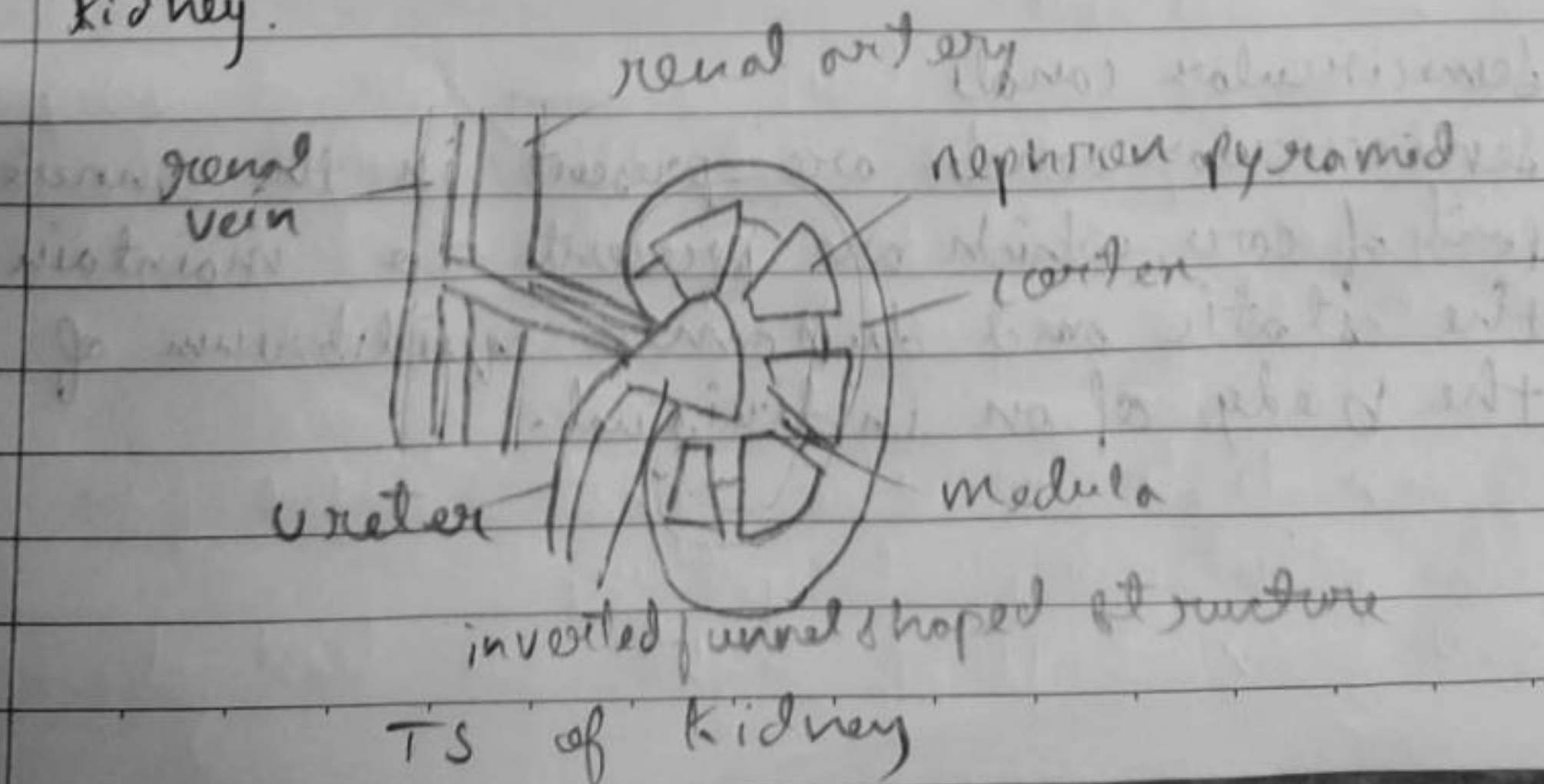
v) a) Auxins
 These are sprayed in order to make the plant to grow properly as it is responsible for apical dominance.

b) Gibberellins
 These are used for stem elongation of plants.

c) Abscissic acid.
 It is used for artificial ripening of fruit.

18. i) Excretion
 It is the removal of nitrogenous wastes by organism in either of the form ammonia, urea or uric acid, in order to get rid from toxic substances, which would cause severe harm to us if allowed to accumulate.

- ii) The substances that are selectively re-absorbed are glucose, essential salts like sodium chloride, iodine, etc. This also includes absorption of water also.
- iii) a) It is the knot of substance different blood vessels that come to kidney for filtration of blood and excrete harmful wastes.
- b) Bowman's capsule is there for the process of ultrafiltration under high pressure. It is referred as ultrafiltration as it is three layer filtration.
- iv) a) PCT
It's function is to absorb all essential substances like that of glucose, salt, like sodium chloride, sodium bicarbonate and iodine.
- b) Loop of henle
It is for re-absorption of water molecules and make the urine concentrated.
- v) Nephron is the structural functional unit of kidney.



19. i) Adrenalline is called flight and fight and fight hormone as this hormone prepares the body to take immediate action against any kind of danger and stressful condition. Thus, it is called emergency hormone, which is released during emergency situations.

ii) Feedback mechanism is the mechanism which controls the secretion of any kind of hormone by any endocrine glands, according to the need of body.

e.g. If basic metabolic rate decreases, pituitary gland ~~makes~~ makes thyroid gland to secrete thyroxine which stabilizes BMR and if this rate goes ~~down~~ ^{up}, it ~~makes~~ pituitary makes thyroid gland to secrete less thyroxine.

iii) a) Thyroid - Thyroxine

b) Pancreas - Insulin and glucagon

iv) a) Iris

Iris * is the structure located in eye which controls the size of pupil according to the external lighting condition.

b) Semicircular canals

Semicircular canals are present in the inner part of ear, which are present to maintain the static and dynamic equilibrium of the body of an individual.

v) a) Hypermetrapia

It is also known as far sightedness, in which a person can see far objects clearly but finds it difficult to see near objects.

b) Synapse

It is the fine gap present between the axon of one neuron and dendrites of other neuron, through which transmission of impulses takes place from one neuron to another.

c) Taste buds

These are the buds present on the tongue, which helps us to understand different kinds of taste. e.g., sweetness, saltiness, sourness, etc.

20. i) Evolution

The structural and functional changes that leads to the improvement of any kind of functioning of body, in a period of time is called evolution.

ii) 2 - Ramapathicus

5 - Cro magnon

iii) Cranial capacity of ramapathicus is 500-600 cc

Cranial capacity of Cro magnon is 1660 cc.

iv) Homo habilis

v) I

- a) In atmosphere, Ozone layer is formed by the combination of 3 vacant oxygen atoms.
- b) Ozone layer is essential for existence of life on earth as it absorbs all ultra violet radiation, which if reaches Earth can cause many diseases to human beings and other animals like skin cancer and blindness.
- c) ~~Reat~~ Release of ozone depleting substance mainly chlorofluorocarbons or freon (CF_2Cl_2) must be reduced.

21. i) Root pressure

It is the hydrostatic pressure developed in the roots which is mainly to move the water in upward direction by in roots, due to concentration of solute in it.

ii) Rate of photosynthesis is low at low temperature because, ~~no~~ much amount of radiant energy is not available.

Rate of photosynthesis is ~~very~~ low at very high temperatures also. This is so because, ~~pho~~. photosynthetic apparatus is destroyed at high temperature (denaturation).

iii) • Green light is the least useful for the plants in photosynthesis.
• The green rays of light are least absorbed.

by the plants, that's why they are not useful for photosynthesis. This is the reason why plants look green.

- iv) Chlorophyll is one of the most important things for photosynthesis. For proving this, we take a variegated plant which has yellow and green patches on its leaves. Yellow part lacks chlorophyll and green parts possess chlorophyll. By detaching plant and keeping it in sunlight and taking its leaves for test, we found that the green patched area turns blue black showing positive starch test, showing that chlorophyll is essential for photosynthesis, yellow patched parts get brown colour, showing negative starch test.

v. a) Purple Inflated : 9
Purple constricted : 3
White Inflated : 3
White constricted : 1

b) Phenotype ratio :- 9:3:3:1

c) Law of independent assortment

This states that in heterozygous condition during formation of gametes, genes assort independently, without blending and influencing each other.

22.) Chromosomes
 Chromosomes are the rod like structures that are formed by constriction and coiling of chromatin network during cell division, which consists of hereditary material.

ii) Mitosis

- Two daughter cells are formed.
- Daughter cells have half some no. of chromosomes as that of parent cells.
- Takes place in body cells.
- This occurs in whole life span.
- For daughter cells are formed.
- Daughter cells have half the no. of chromosomes as that of parent cells.
- Takes place in formation of gametes.
- This occurs only in reproductively active age.

iii) Osmosis is the process being studied.

Root hairs are of immense importance for plants which provides extensive surface area for absorption of water and minerals from soil.

iv) External factors that affects the rate of transpiration are:

- Intensity of light
- ~~Had~~ Humidity of the atmosphere

- | | Parts | Functions |
|----|---------|--|
| 1. | Ovary | Produces female gamete ova |
| 2. | Oviduct | Carries the ova to the uterus |
| 3. | Vagina | It is the structure in which embryo develops and grows |